

Haowen Hu, M.S.
Ph.D. Candidate

Department of Animal Science
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CAREER OBJECTIVE

To become an independent researcher who leads interdisciplinary work at the intersection of animal agriculture, sustainability, data science, and decision support, developing practical tools that improve environmental performance, productivity, and resilience in livestock systems.

EDUCATION

Ph.D.	Cornell University, <i>Animal Science</i> ; Minor in <i>Systems Engineering</i> GPA: 4.07/4.30 Advisors: Drs. Kristan F. Reed and Thomas R. Overton	Expected December 2026
M.S.	University of Wisconsin-Madison, <i>Dairy Science</i> GPA: 4.00/4.00 Advisor: Dr. Sebastian Arriola-Apelo	August 2020
B.S.	China Agricultural University, <i>Animal Science</i> GPA: 3.90/4.00 Cornell University, <i>Animal Science</i> GPA: 3.93/4.30 Honors Thesis Advisor: Dr. Michael Van Amburgh	November 2018 May 2018

RESEARCH INTERESTS

Dairy sustainability; whole-farm environmental and systems modeling; digital twins for livestock systems; greenhouse gas and ammonia emissions quantification; sensor-based monitoring; data integration; AI-assisted agricultural modeling, including LLMs and agent-based workflows.

TECHNICAL SKILLS

Programming, Statistics & Data Analysis: Python, R, Excel VBA; statistical and machine learning modeling; uncertainty and sensitivity analysis; Git version control; SQL databases
Systems Modeling, AI & Data Workflows: RuFaS and COMET-Farm whole-farm modeling; Monte Carlo simulation; Markov chain simulation; data assimilation; LLM-assisted extraction; AI agents (LangGraph)
Field Methods & Sensors: GreenFeed systems and gas emissions data processing; sensor-based monitoring; ventilation-based flux estimation; manure and biological sampling
Research Skills: scientific writing; literature synthesis; systematic review

SELECTED PUBLICATIONS AND ABSTRACTS

Peer-Reviewed Journal Articles

1. **Hu, H.**, Oliver, J. P., Niu, P., Leytem, A. B., & Reed, K. F. 2026. Assessing precision and accuracy of manure volatile solids, total nitrogen, and total ammoniacal nitrogen analyses across 6 commercial laboratories. *Journal of Dairy Science Communications*, 7(4), 487–493. <https://doi.org/10.3168/jdsc.2025-1007>.
2. Sharma, S., Liu, E., van Leerdam, M., **Hu, H.**, Villalobos-Barquero, R., Dorea, J. R. R., Cabrera, V. E., Iwersen, M., Bewley, J. M., Soyeurt, H., & Hostens, M. 2026. Invited review: Milking the data for value-driven dairy farming. *Journal of Dairy Science*, 109(4), 3137–3157. <https://doi.org/10.3168/jds.2025-27011>.

3. Reed, K. F., Tricarico, J. M., HekmatiAthar, S., Waddell, J. S., Andreen, D. A., Briggs, K. R., Liu, A., Tomlinson, N. D., Adamchick, J., Cabrera, V. E., **Hu, H.**, Gong, Y., Graef, G. M., Villalobos-Barquero, M. R., Oliver, J. P., Nydam, D. V., Ayache, N., & McClintock, L. 2025. Invited review: The Ruminant Farm Systems (RuFaS) model is a platform to support future research and actions for sustainable dairy farming. *Journal of Dairy Science Communications*, 6(Suppl. 1), S15–S22. <https://doi.org/10.3168/jdsc.2025-0861>.
4. **Hu, H.**, Whitcomb, C. A., Ploetz, T. E., & Reed, K. F. 2025. Transdisciplinary model-based systems engineering in the development of the Ruminant Farm Systems model. *Frontiers in Sustainability*, 6, 1561453. <https://doi.org/10.3389/frsus.2025.1561453>.
5. Gong, Y., **Hu, H.**, Reed, K. F., & Cabrera, V. E. 2025. Advancing dairy farm simulations: A 2-step approach for tailored lactation curve estimation and its systemic impacts. *Journal of Dairy Science*, 108(9), 9681–9695. <https://doi.org/10.3168/jds.2025-26334>.
6. Cabrera, V. E., Bewley, J., Breunig, M., Breunig, T., Cooley, W., De Vries, A., Fourdraine, R., Giordano, J. O., Gong, Y., Greenfield, R., **Hu, H.**, Lenkaitis, A., Niu, M., Noronha, E. A. F., & Sullivan, M. 2025. Data integration and analytics in the dairy industry: Challenges and pathways forward. *Animals*, 15(3), 329. <https://doi.org/10.3390/ani15030329>.

Manuscripts In Press, Under Review, and In Preparation

7. Qian, C., Jung, Y., **Hu, H.**, Gong, Y., Negreiro, A. N., Cabrera, V. E., Ivanek, R., & Wiedmann, M. 2026. Future-proofing agricultural research: FAIR principles for Agriculture AI Agents (FAIR4AG2). *Frontiers in Artificial Intelligence – AI in Food, Agriculture and Water*. Under review.
8. **Hu, H.**, Briggs, K. R., Adamchick, J., Andreen, D. A., Waddell, J. S., HekmatiAthar, S., Gong, Y., Graef, G. M., Villalobos-Barquero, M. R., Ivanek, R., Whitcomb, C. A., Cabrera, V. E., Overton, T. R., Tricarico, J. M., & Reed, K. F. A methodological comparison of COMET-Farm and RuFaS in estimating greenhouse gas emissions from animal and manure management on 5 US Holstein dairies. In preparation. Planned submission: Aug 2026.

Conference Abstracts and Proceedings

9. **Hu, H.**, Ge, X., Zhu, R., Fang, N., Cabrera, V. E., & Reed, K. F. 2026. Markov-chain herd simulation quantifies the economic feasibility of 3-nitrooxypropanol and bromoform-containing additives for enteric methane mitigation. *Animal Science Modellers' Group Annual Meeting*, Milwaukee, WI. Presented.
10. **Hu, H.**, Perez, M., Oliver, J., Jacome, A., Scattolini, F., Giordano, J., & Reed, K. 2026. High-temporal-resolution ammonia concentration measurements in a naturally ventilated dairy barn to inform RuFaS housing ammonia prediction refinement. *European Geosciences Union General Assembly*, Vienna, Austria. <https://doi.org/10.5194/egusphere-egu26-8438>. Presented.
11. **Hu, H.**, Xu, H., Luo, Y., & Reed, K. F. 2025. Evaluating the RuFaS model: Insights into NH₃ emission simulation via data assimilation. *American Dairy Science Association Annual Meeting*, Louisville, KY. Presented.
12. **Hu, H.**, Waddell, J., Cabezas-Garcia, E. H., & Reed, K. F. 2024. Automated ration formulation for lactating cows within the Ruminant Farm Systems model results in different diets and emissions estimates when using different nutrient requirement models. *Canadian Journal of Animal Science*, 104(Suppl. 1), S1–S7. <https://doi.org/10.1139/cjas-2024-0024#subarticle-8>. Presented.

HONORS AND AWARDS

National Dairy Leadership Scholarship (\$3,000) – <i>NMPF</i>	2026
Cornell CALS Animal Science Grad Travel Award (\$300) – <i>Cornell University</i>	2025 – 2026
Cornell Graduate School Travel Grant (\$400–\$700) – <i>Cornell University</i>	2023 – 2026

National Science Foundation I-Corps Regional Program (\$5,000) – NSF	2025
Green Technology Innovation Fellowship – Cornell University	2025
American Dairy Science Association Travel Grant Award (\$500) – ADSA	2025
Third Place – AI Hackathon for Livestock Data (Remote)	2024
Full Scholarship – São Paulo School of Advanced Science on Precision Livestock Farming (Brazil)	2024
ADSA 46 th Discover Conference Graduate Student Assistant – ADSA	2024
Danny G. Fox Graduate Fellowship – Cornell University	2023
Foundation for Food & Agriculture Research Fellowship (\$15,000) – FFAR	2023

RESEARCH FUNDING SUPPORT

USDA NIFA CAST Award (#2023-77038-38865). 2023

The Farm of the Future: Harnessing Data-Driven Technology for a Sustainable and Resilient US Agricultural System.

PI: Julio Giordano; Project supervisor: Kristan F. Reed.

Role: Graduate Research Assistant / project contributor.

USDA NIFA IDEAS Award (#2020-68014-31466). 2020

Ruminant Farm Systems (RuFaS).

PI: Kristan F. Reed.

Role: Graduate Research Assistant / project contributor.

RELEVANT RESEARCH / WORK EXPERIENCE

Cornell University, Ithaca, NY January 2023 – Present

Graduate Research Assistant

- Leading field quantification of barn ammonia emissions using sensor networks and ventilation-based flux methods.
- Integrating field measurements with modeling via data assimilation to refine ammonia emission parameters in RuFaS.
- Simulated 12 US dairy systems in RuFaS and benchmarked predictions against published data.
- Conducted RuFaS–COMET-Farm comparison to identify methodological drivers of greenhouse gas estimates.
- Contributed to development of RuFaS as the modeling engine for FARM Environmental Stewardship Program Version 3, supporting industry-aligned sustainability assessment.
- Directed multi-lab evaluation of manure nutrient analyses (6 commercial labs) to quantify measurement variability.
- Designed stakeholder-informed use cases and system requirements to improve decision-support tool usability.
- Mentored 4 undergraduate students and co-supervised 1 senior honors thesis.

C-Lock Inc., Rapid City, SD July 2020 – November 2022

Lead Data Analyst (April 2022 – November 2022)

Junior Data Analyst (July 2020 – April 2022)

- Trained and supervised 3 junior analysts, led code reviews (Excel VBA, C++), and represented the organization at field deployments and technical events.
- Developed and standardized data workflows and resolved technical issues to support efficient data processing.

- Processed and validated ruminant gas emissions data using GreenFeed systems for environmental assessments.

TEACHING EXPERIENCE

Colgate University, Hamilton, NY

December 2024

Invited Guest Lecturer

- Delivered a guest lecture in *CORE 135 FY FSEM Renewable Energy* to 20 undergraduate students on sustainable dairy farming and an introduction to the RuFaS model.

Cornell University, Ithaca, NY

August 2023 – Present

Graduate Teaching Assistant / Guest Lecturer

- Co-delivered a guest lecture in *ANSC 4040/6040 Data Science Applications in Agriculture* on the RuFaS model and guided 8 students through dairy farm simulation as a decision-support tool.
- Delivered a guest lecture in *ANSC 1205 Current Dairy Topics Explored through Writing* on conducting literature reviews to support evidence-based public commentary for 27 undergraduate students.
- Supported instruction and assessment for *ANSC 1205 Current Dairy Topics Explored through Writing*, *ANSC 3510/5510 Dairy Herd Management*, and *ANSC 3500 Meat* across semester-long course sections.

LEADERSHIP / SERVICE / PROFESSIONAL ACTIVITIES

Green Technology Innovation Fellows Program, Ithaca, NY

May 2025 – May 2026

Fellow

- Conducted 50+ customer interviews with dairy farmers, advisors, and other industry stakeholders to identify barriers and opportunities in farm decision-making.
- Selected to attend the American Farm Bureau Convention in Anaheim, CA, and conducted 30+ stakeholder interviews with farmers, government representatives, and other agriculture professionals.
- Delivered 6 presentations on customer discovery findings and the development of decision-support solutions for agriculture.

Foundation for Food & Agriculture Research, Raleigh, NC

August 2023 – May 2026

Fellow

- Completed 150+ hours of professional development workshops in communication, leadership, negotiation, scientific communication, entrepreneurship, and project management.
- Selected as 1 of 30 fellows through a highly competitive national application process.

Cornell Animal Science Graduate Student Association, Ithaca, NY

May 2024 – May 2025

Professional Development and Operations Manager

- Organized 8 professional development events for a graduate community of about 40, with each event attended by 25 to 35 students.
 - Arranged the Cornell Women in Animal Science Panel to promote diversity in the livestock industry.
 - Coordinated the 2nd Sciencing Challenge at Cornell University to support student innovation and entrepreneurial thinking.
 - Organized a public speaking workshop in collaboration with Toastmasters.